

torch.lu_solve#

https://pytorch.org/docs/stable/generated/torch.lu_solve.html

Description:

Returns the LU solve of the linear system $Ax=b$ using the partially pivoted LU factorization of A from `lu_factor()`. This function supports float, double, cfloat and cdouble dtypes for input. `torch.lu_solve()` is deprecated in favor of `torch.linalg.lu_solve()`. `torch.lu_solve()` will be removed in a future PyTorch release. `X = torch.lu_solve(B, LU, pivots)` should be replaced with `b (Tensor)` – the RHS tensor of size $(*,m,k)$, where $*$ is zero or more batch dimensions.

Function Signature

`torch.lu_solve(b, LU_data, LU_pivots, *, out=None) → Tensor#`

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
X = linalg.lu_solve(LU, pivots, B)
```

```
>>> A = torch.randn(2, 3, 3)
>>> b = torch.randn(2, 3, 1)
>>> LU, pivots = torch.linalg.lu_factor(A)
>>> x = torch.lu_solve(b, LU, pivots)
>>> torch.dist(A @ x, b)
tensor(1.00000e-07 *
      2.8312)
```

Notes & Warnings

WARNING

Warning `torch.lu_solve()` is deprecated in favor of `torch.linalg.lu_solve()`. `torch.lu_solve()` will be removed in a future PyTorch release. `X = torch.lu_solve(B, LU, pivots)` should be replaced with `X = linalg.lu_solve(LU, pivots, B)`

Detailed Documentation

Documentation

Returns the LU solve of the linear system $Ax=b$ using the partially pivoted LU factorization of A from `lu_factor()`.

This function supports float, double, cfloat and cdouble dtypes for input.

`torch.lu_solve()` is deprecated in favor of `torch.linalg.lu_solve()`. `torch.lu_solve()` will be removed in a future PyTorch release. `X = torch.lu_solve(B, LU, pivots)` should be replaced with

b (Tensor) – the RHS tensor of size $(\ast, m, k)$, where $\ast$ is zero or more batch dimensions.

LU_data (Tensor) – the pivoted LU factorization of A from `lu_factor()` of size $(\ast, m, m)$, where $\ast$ is zero or more batch dimensions.

LU_pivots (IntTensor) – the pivots of the LU factorization from `lu_factor()` of size $(\ast, m)$, where $\ast$ is zero or more batch dimensions. The batch dimensions of `LU_pivots` must be equal to the batch dimensions of `LU_data`.

out (Tensor, optional) – the output tensor.